

Version comparison: EXHALE v1.0 vs current (He/H + metal diffusion)

on HD 209458b and WASP-121b

Status note. The numbers in this memo were measured against the trace helium kernel, which transported the ratio $f = n_{\text{He}}/n_{\text{H}}$ under a cap $f \leq (\text{He}/\text{H})_{\text{base}}$. That kernel has since been replaced by `src/modules/functions/binary_element_diffusion.f90`, which transports the helium MASS FRACTION $X = \rho_{\text{He}}/\rho$ of a binary H/He mixture, carries molecular carriers and stage-resolved friction, and has no cap and no base pile-up limiter (its M-matrix keeps $0 \leq X \leq 1$ per step without clipping). The measurements below are kept as a record of what the trace kernel gave; they have not been re-run under the binary operator, so read the two regimes and the sign of each effect, not the digits. Design of the current operator: `docs/binary_diffusion_design.md`. The trace treatment of the metals is carried over unchanged into the new module.

1. Purpose and setup

Quantify what the new diffusive separation changes in EXHALE's predictions, by running the *same* planet on two code versions, for *two contrasting planets*:

- **v1.0** (ATES/EXHALE_v1.0/): clean GitHub main + the temperature-dependent Penning rate. *No* diffusive separation.
- **current** (EXHALE_v1.00/): v1.0 physics + He/H and element-by-element metal diffusion (He_diffusion + He_metal_diffusion, ambipolar on).

Both share the temperature-dependent Penning rate, so the only difference is the diffusion. Setup: He ²³S on, trace metals from `metals.inp` (HD 209458b: C/N/O; WASP-121b: C/N/O/Mg/Ca/Na/Fe), identical `input.inp` per planet, 15000 relaxation steps. Runs/figures/script: `docs/version_compare/`.

2. Headline numbers

Two regimes. On the gentler HD 209458b the escape is slow enough that *helium itself* diffusively separates, cutting the He I 10830 line by $\sim 2.6\times$, and the heavier C/N/O deplete more than He aloft ($\text{He} > \text{C} > \text{N} > \text{O}$ at $3 R_p$). On the furious WASP-121b (~ 4 dex stronger wind) advection dominates settling for *every* species: helium *and* the metals are dragged out essentially unfractionated (10830 unchanged; metal/H tracks He/H). This is the (settling speed)/(wind speed) scaling of Koskinen et al. (2013) / Xing et al. (2023): separation is suppressed by fast escape and, where it operates at all, is stronger for heavier species.

Correction note: an earlier version showed WASP-121b metals collapsing to zero by $\sim 1.25 R_p$ (a “metal homopause”). That was an artifact of two bugs in the metal rescale (an $r_X \leq 1$ clamp that made depletion a one-way ratchet, and a skip of exhausted cells that made holes permanent), which locked in transient early-relaxation settling before the wind

HD 209458b (hot Jupiter, $\log_{10} \dot{M} \approx 9.3$: gentler escape)

quantity	v1.0 (no diff)	current (diff)	change
He/H at $3 R_p$	0.083 (flat)	0.014	−83%
peak $n(\text{He } 2^3\text{S})$ [cm^{-3}]	177	102	−42%
He I 10830 line-center absorption	70.5%	27.1%	−2.6×
$\log_{10} \dot{M}$ [g s^{-1}]	9.31	9.04	−0.27 dex

WASP-121b (ultra-hot Jupiter, $\log_{10} \dot{M} \approx 13.3$: furious escape)

quantity	v1.0 (no diff)	current (diff)	change
He/H (min, $1.05\text{--}1.5 R_p$)	0.083	0.053	−36% (weaker)
peak $n(\text{He } 2^3\text{S})$ [cm^{-3}]	3517	2383	−32%
He I 10830 line-center absorption	53.7%	53.6%	~unchanged
metal/H (elem/base)	1.0	tracks He ($\sim 0.6\text{--}1.0$; no extra fractionation)	none
$\log_{10} \dot{M}$ [g s^{-1}]	13.34	13.32	unchanged

developed. With the fix, metals recover with the wind and track He, and the v1-vs-v2 temperature structures agree; the earlier “ $\sim 2500\text{ K}$ hotter” claim is also retracted.

3. HD 209458b

HD 209458b: EXHALE v1.0 (no diffusion) vs current (He/H + metal diffusion)

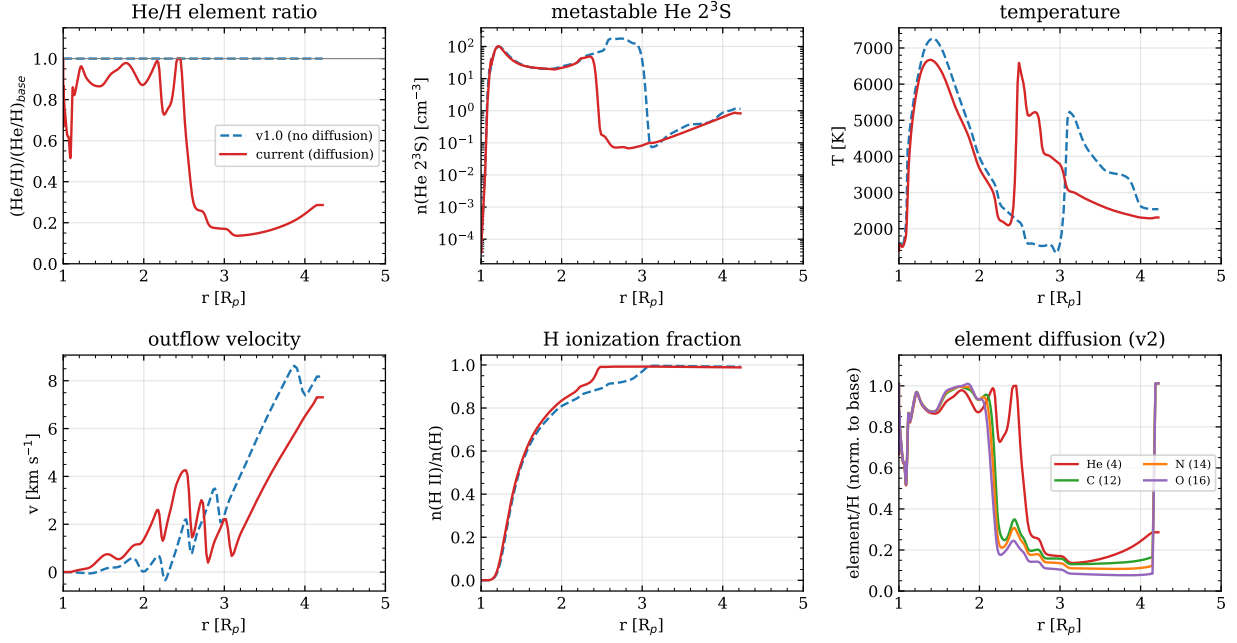


Figure 1: HD 209458b, v1.0 (blue dashed) vs current (red). He/H falls to $\sim 0.15\text{--}0.3\times$ aloft; the $\text{He } 2^3\text{S}$ reservoir contracts. Bottom-right: C/N/O deplete more than He aloft (mass ordering $\text{He} > \text{C} > \text{N} > \text{O}$ above $\sim 2.5 R_p$).

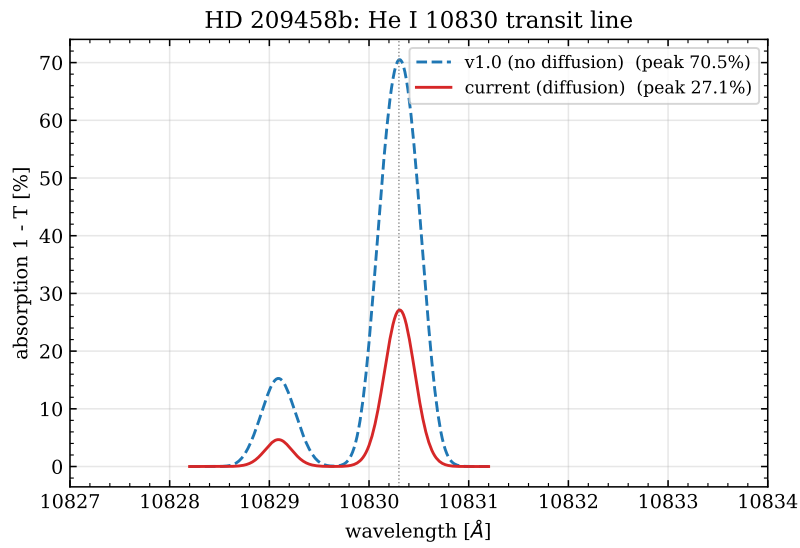


Figure 2: He I 10830 line: diffusion lowers the line-center absorption $70.5\% \rightarrow 27.1\%$ and narrows the profile.

4. WASP-121b

WASP-121b: EXHALE v1.0 (no diffusion) vs current (He/H + metal diffusion)

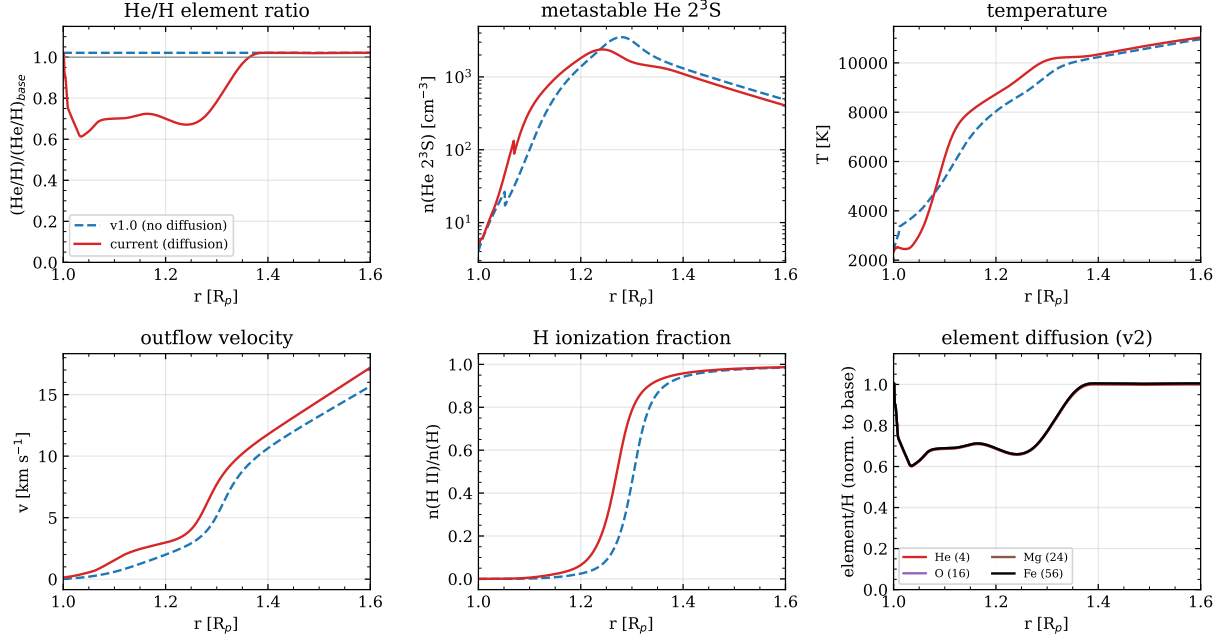


Figure 3: WASP-121b (compact domain, escape radius $1.5 R_p$). He/H separates only modestly (~ 0.6 – $0.75\times$, returning to ~ 1 above $1.4 R_p$) because the ~ 4 -dex-stronger wind drags helium out. The metals (O, Mg, Fe) overlap He almost exactly: even Fe (mass 56) is advection-dominated here, so no additional fractionation develops. The v1/v2 temperature and velocity structures are nearly identical.

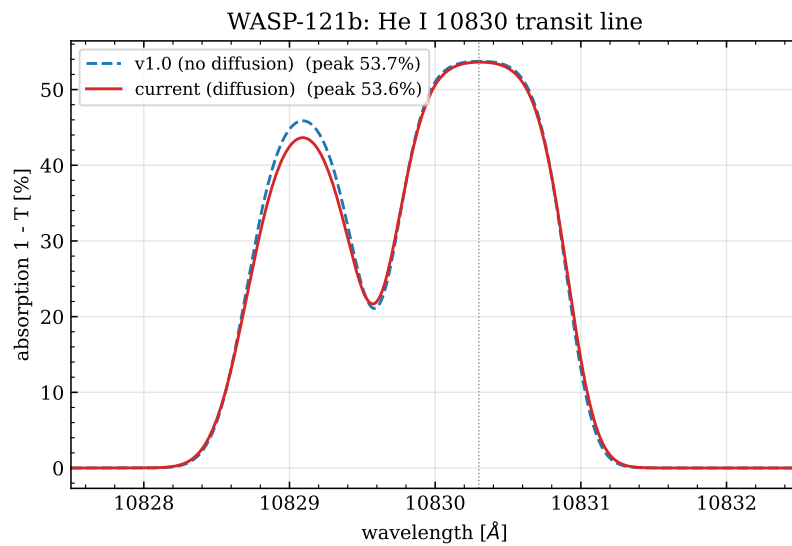


Figure 4: He I 10830 line: the core is essentially unchanged (53.7%→53.6%) since helium is not strongly separated in the line-forming region.

5. Interpretation and caveats

Interpretation.

- **He/H.** On HD 209458b settling competes with the slow wind, so He/H declines with altitude and 10830 weakens; on WASP-121b the wind is ~ 4 dex stronger, so helium is dragged out nearly unfractionated.
- **He 2^3S / 10830.** The metastable tracks He abundance: weakened where He is depleted (HD 209458b), preserved where He stays mixed (WASP-121b).
- **Metals.** Each trace metal diffuses independently with its own mass, binary D , and ambipolar-corrected settling. Where diffusion competes with the wind (HD 209458b aloft) heavier elements deplete faster ($\text{He} > \text{C} > \text{N} > \text{O}$); where advection dominates (WASP-121b) even Fe stays locked to He. Relevant for metal-line transmission (Mg II/Ca II/Na I in EXHALE_transit.py).

Caveats.

- **Absolute values uncalibrated.** The 27–70% He 10830 depths far exceed observations ($\sim 1\text{--}2\%$); heating efficiency / XUV / He 2^3S microphysics were not tuned. The *relative* change and the two-planet *contrast* are the robust results.
- **Scheme / convergence sensitivity.** Step-capped (not Newton-finished), so $\dot{M}$ is indicative; the He-separation magnitude is somewhat sensitive to the settling discretization (a Peclet-hybrid central/upwind scheme: central where well-resolved, upwind for the stiff heavy-metal settling).
- **Phase-1/2 numerics.** Mild non-monotonic structure from the He/H cap and the not-fully-developed wind; see design_hehe_diffusion.md §7d–§7e. Diffusion flags **default OFF**. *The cap is gone in the current operator (status note); the scalar He_Kzz now defaults to 0 rather than $10^9 \text{ cm}^2 \text{ s}^{-1}$, and a Lower atmosphere profile may supply $K_{zz}(p)$ per cell instead.*